HVAC System Design

Design study. Numerical execution is not full engineering or regulatory acceptance. All requirement states below are retained from the verified workflow.

Source-classified load basis

Input	Value	Classification	Source
area	1022 m2	given	Original layout drawing area label; interpreted net geometry differs
cooling_allowance	189 W/m2	given	Canvas BSE3227 Space air diffusion design and air duct sizing.pdf p1; SHA256 210d482549a3fbab5b4aef0db7c159cec9d36eae336a1e4ca64f3b30f3a9acf0; preliminary office allowance / approximate SHF, not detailed room load
sensible_fraction	0.9 1	given	Canvas BSE3227 Space air diffusion design and air duct sizing.pdf p1; SHA256 210d482549a3fbab5b4aef0db7c159cec9d36eae336a1e4ca64f3b30f3a9acf0; preliminary office allowance / approximate SHF, not detailed room load
density	1.2 kg/m3	given	Canvas lecture Space air diffusion design and air duct sizing.pdf p9 friction-chart condition: rho=1.20 kg/m3; adopted constant-property trial air density
specific_heat	1006 J/(kg K)	assumed	Explicit trial air-property/temperature assumption
temperature_difference	12 K	assumed	Explicit trial air-property/temperature assumption

Planar metres; interpreted net office excluding source columns; semantic door closures and original scale remain under review. Source office-obstacles SHA256 879687f901ee7095ce9939354201cf3219ccb989dd1b7e95e0ab99b53dda48e1;Declared common-right-column alternative,24 chosen position changes;not measured source locations. Original CAD remains preserved.

Cooling and supply airflow

Total thermal cooling kW	Trial sensible thermal kW	Trial supply L/s
193.158	173.842	12000.4

Displayed thermal loads use three decimal places in kW and airflow uses one decimal place in L/s. Calculations retain their original precision in the source artifacts.

A uniform allowance and sensible fraction do not establish the actual room sensible load for ADPI.

ADPI applicability and evidence

Applicability: pending. 640 maximum/minimum directional cases. This is source-scoped ratio screening;no measured or predicted ADPI is claimed.

Selected criterion: Canvas BSE3227 original lecture p5; ASHRAE Fundamentals2005 table reproduced in class. Room sensible-load range: 35 to 160 W/m2. T0.25/L range: 1 to 3.4.

Supplied room-load value: 170.1 W/m2. Within source range: False. An unverified preliminary value does not establish the actual room load or an ADPI failure.

Missing applicability facts: Room sensible load, Characteristic lengths, Installation conditions. Rejected facts: None declared.

Screened outcomes: not_evaluated=640, within_sourced_ratio_range=0, fail=0. All ratios within the selected applicable scope: False. Pending applicability is not a comfort pass or a measured failure.

Room sensible load: Unverified.

Product category within selected criterion: Declared supported.

Characteristic lengths: Unverified.

Installation conditions: Unverified.

Criterion edition selection: Declared supported.

Evidence for Product category within selected criterion 1: Space air diffusion design and air duct sizing.pdf; p5 classroom example and2005 selection table. Explicit course-specific1400-family mapping;retains classroom edition and stated bounds

Evidence for Product category within selected criterion 2: krueger-introduction.pdf; B1-4 introduction and family heading. Square-face,four-cone,round-neck louvered family;not a measured ADPI or later handbook category mapping

Evidence for Product category within selected criterion 3: category-review.json; findings and retained_unknowns. Recorded interpretation and retained limits;reviewer assertion,not independent authentication

Evidence for Product category within selected criterion 4: classroom-table.png; Original lecture p5 inspected table render. Copied hash-identical historical visual evidence;does not clear room load,L or installation unknowns

Evidence for Product category within selected criterion 5: krueger-introduction.png; Manufacturer introduction inspected render. Copied hash-identical historical visual evidence;does not clear room load,L or installation unknowns

Source hashes and exact evidence notes are retained in the report manifest. Declarations and source-byte identity do not authenticate physical conditions or establish comfort/engineering acceptance.

Maximum and minimum terminal lookup

Declared common-right-column layout alternative;flows inherit fresh source-classified allocation. Cardinal polygon/aligned-midpoint ray lengths are geometric diagnostics,not authenticated ASHRAE characteristic lengths. Room load,wall/column/jet interpretation and installation applicability remain pending.

Listed L values remain input assumptions pending characteristic-length applicability review. No ADPI pass is inferred.

ID	Direction	Max L/s	Min L/s	Input L m	Max throw m	Min throw m	Max T/L	Min T/L
SD-01	east	132.2	52.9	1.860	4.203	2.344	2.259	1.260
SD-01	west	132.2	52.9	1.860	4.203	2.344	2.259	1.260
SD-01	north	132.2	52.9	1.708	4.203	2.344	2.461	1.372
SD-01	south	132.2	52.9	1.707	4.203	2.344	2.461	1.372
SD-02	east	147.8	59.1	1.860	4.392	2.609	2.361	1.402
SD-02	west	147.8	59.1	1.860	4.392	2.609	2.361	1.402
SD-02	north	147.8	59.1	1.708	4.392	2.609	2.572	1.528
SD-02	south	147.8	59.1	1.707	4.392	2.609	2.572	1.528
SD-03	east	147.8	59.1	1.860	4.392	2.609	2.361	1.402
SD-03	west	147.8	59.1	1.860	4.392	2.609	2.361	1.402
SD-03	north	147.8	59.1	1.708	4.392	2.609	2.572	1.528
SD-03	south	147.8	59.1	1.707	4.392	2.609	2.572	1.528
SD-04	east	141.0	56.4	1.860	4.309	2.492	2.317	1.340
SD-04	west	141.0	56.4	1.860	4.309	2.492	2.317	1.340
SD-04	north	141.0	56.4	1.708	4.309	2.492	2.523	1.460
SD-04	south	141.0	56.4	1.707	4.309	2.492	2.523	1.460

ID	Direction	Max L/s	Min L/s	Input L m	Max throw m	Min throw m	Max T/L	Min T/L
SD-05	east	142.3	56.9	1.860	4.325	2.515	2.325	1.352
SD-05	west	142.3	56.9	1.860	4.325	2.515	2.325	1.352
SD-05	north	142.3	56.9	1.708	4.325	2.515	2.533	1.473
SD-05	south	142.3	56.9	1.707	4.325	2.515	2.533	1.473
SD-06	east	147.8	59.1	1.860	4.392	2.609	2.361	1.402
SD-06	west	147.8	59.1	1.860	4.392	2.609	2.361	1.402
SD-06	north	147.8	59.1	1.708	4.392	2.609	2.572	1.528
SD-06	south	147.8	59.1	1.707	4.392	2.609	2.572	1.528
SD-07	east	147.8	59.1	1.860	4.392	2.609	2.361	1.402
SD-07	west	147.8	59.1	1.860	4.392	2.609	2.361	1.402
SD-07	north	147.8	59.1	1.708	4.392	2.609	2.572	1.528
SD-07	south	147.8	59.1	1.707	4.392	2.609	2.572	1.528
SD-08	east	157.2	62.9	2.264	4.506	2.768	1.990	1.223
SD-08	west	157.2	62.9	1.860	4.506	2.768	2.422	1.488
SD-08	north	157.2	62.9	1.708	4.506	2.768	2.639	1.621
SD-08	south	157.2	62.9	1.707	4.506	2.768	2.639	1.621
SD-09	east	153.6	61.4	1.745	4.462	2.706	2.557	1.551
SD-09	west	153.6	61.4	2.264	4.462	2.706	1.971	1.195
SD-09	north	153.6	61.4	1.708	4.462	2.706	2.613	1.585
SD-09	south	153.6	61.4	1.707	4.462	2.706	2.613	1.585
SD-10	east	138.6	55.5	1.745	4.281	2.453	2.454	1.406
SD-10	west	138.6	55.5	1.745	4.281	2.453	2.454	1.406
SD-10	north	138.6	55.5	1.708	4.281	2.453	2.507	1.437
SD-10	south	138.6	55.5	1.707	4.281	2.453	2.507	1.437
SD-11	east	138.6	55.5	1.745	4.281	2.453	2.454	1.406
SD-11	west	138.6	55.5	1.745	4.281	2.453	2.454	1.406

ID	Direction	Max L/s	Min L/s	Input L m	Max throw m	Min throw m	Max T/L	Min T/L
SD-11	north	138.6	55.5	1.708	4.281	2.453	2.507	1.437
SD-11	south	138.6	55.5	1.707	4.281	2.453	2.507	1.437
SD-12	east	123.0	49.2	1.745	4.037	2.188	2.314	1.254
SD-12	west	123.0	49.2	1.745	4.037	2.188	2.314	1.254
SD-12	north	123.0	49.2	1.708	4.037	2.188	2.364	1.281
SD-12	south	123.0	49.2	1.707	4.037	2.188	2.364	1.281
SD-13	east	147.8	59.1	1.860	4.392	2.609	2.361	1.402
SD-13	west	147.8	59.1	1.860	4.392	2.609	2.361	1.402
SD-13	north	147.8	59.1	1.707	4.392	2.609	2.572	1.528
SD-13	south	147.8	59.1	1.708	4.392	2.609	2.572	1.528
SD-14	east	147.8	59.1	1.860	4.392	2.609	2.361	1.402
SD-14	west	147.8	59.1	1.860	4.392	2.609	2.361	1.402
SD-14	north	147.8	59.1	1.707	4.392	2.609	2.572	1.528
SD-14	south	147.8	59.1	1.708	4.392	2.609	2.572	1.528
SD-15	east	147.8	59.1	1.860	4.392	2.609	2.361	1.402
SD-15	west	147.8	59.1	1.860	4.392	2.609	2.361	1.402
SD-15	north	147.8	59.1	1.707	4.392	2.609	2.572	1.528
SD-15	south	147.8	59.1	1.708	4.392	2.609	2.572	1.528
SD-16	east	147.8	59.1	1.860	4.392	2.609	2.361	1.402
SD-16	west	147.8	59.1	1.860	4.392	2.609	2.361	1.402
SD-16	north	147.8	59.1	1.707	4.392	2.609	2.572	1.528
SD-16	south	147.8	59.1	1.708	4.392	2.609	2.572	1.528
SD-17	east	147.8	59.1	1.860	4.392	2.609	2.361	1.402
SD-17	west	147.8	59.1	1.860	4.392	2.609	2.361	1.402
SD-17	north	147.8	59.1	1.707	4.392	2.609	2.572	1.528
SD-17	south	147.8	59.1	1.708	4.392	2.609	2.572	1.528

ID	Direction	Max L/s	Min L/s	Input L m	Max throw m	Min throw m	Max T/L	Min T/L
SD-18	east	147.8	59.1	1.860	4.392	2.609	2.361	1.402
SD-18	west	147.8	59.1	1.860	4.392	2.609	2.361	1.402
SD-18	north	147.8	59.1	1.707	4.392	2.609	2.572	1.528
SD-18	south	147.8	59.1	1.708	4.392	2.609	2.572	1.528
SD-19	east	147.8	59.1	1.860	4.392	2.609	2.361	1.402
SD-19	west	147.8	59.1	1.860	4.392	2.609	2.361	1.402
SD-19	north	147.8	59.1	1.707	4.392	2.609	2.572	1.528
SD-19	south	147.8	59.1	1.708	4.392	2.609	2.572	1.528
SD-20	east	163.9	65.5	2.264	4.586	2.881	2.026	1.273
SD-20	west	163.9	65.5	1.860	4.586	2.881	2.466	1.549
SD-20	north	163.9	65.5	1.707	4.586	2.881	2.686	1.687
SD-20	south	163.9	65.5	1.708	4.586	2.881	2.686	1.687
SD-21	east	159.3	63.7	1.745	4.531	2.803	2.597	1.607
SD-21	west	159.3	63.7	2.264	4.531	2.803	2.001	1.238
SD-21	north	159.3	63.7	1.707	4.531	2.803	2.653	1.642
SD-21	south	159.3	63.7	1.708	4.531	2.803	2.653	1.642
SD-22	east	138.6	55.5	1.745	4.281	2.453	2.454	1.406
SD-22	west	138.6	55.5	1.745	4.281	2.453	2.454	1.406
SD-22	north	138.6	55.5	1.707	4.281	2.453	2.507	1.437
SD-22	south	138.6	55.5	1.708	4.281	2.453	2.507	1.437
SD-23	east	138.6	55.5	1.745	4.281	2.453	2.454	1.406
SD-23	west	138.6	55.5	1.745	4.281	2.453	2.454	1.406
SD-23	north	138.6	55.5	1.707	4.281	2.453	2.507	1.437
SD-23	south	138.6	55.5	1.708	4.281	2.453	2.507	1.437
SD-24	east	138.6	55.5	1.745	4.281	2.453	2.454	1.406
SD-24	west	138.6	55.5	1.745	4.281	2.453	2.454	1.406

ID	Direction	Max L/s	Min L/s	Input L m	Max throw m	Min throw m	Max T/L	Min T/L
SD-24	north	138.6	55.5	1.707	4.281	2.453	2.507	1.437
SD-24	south	138.6	55.5	1.708	4.281	2.453	2.507	1.437
SD-25	east	147.8	59.1	1.860	4.392	2.609	2.361	1.402
SD-25	west	147.8	59.1	1.860	4.392	2.609	2.361	1.402
SD-25	north	147.8	59.1	1.707	4.392	2.609	2.572	1.528
SD-25	south	147.8	59.1	1.707	4.392	2.609	2.572	1.528
SD-26	east	147.8	59.1	1.860	4.392	2.609	2.361	1.402
SD-26	west	147.8	59.1	1.860	4.392	2.609	2.361	1.402
SD-26	north	147.8	59.1	1.707	4.392	2.609	2.572	1.528
SD-26	south	147.8	59.1	1.707	4.392	2.609	2.572	1.528
SD-27	east	147.8	59.1	1.860	4.392	2.609	2.361	1.402
SD-27	west	147.8	59.1	1.860	4.392	2.609	2.361	1.402
SD-27	north	147.8	59.1	1.707	4.392	2.609	2.572	1.528
SD-27	south	147.8	59.1	1.707	4.392	2.609	2.572	1.528
SD-28	east	147.8	59.1	1.860	4.392	2.609	2.361	1.402
SD-28	west	147.8	59.1	1.860	4.392	2.609	2.361	1.402
SD-28	north	147.8	59.1	1.707	4.392	2.609	2.572	1.528
SD-28	south	147.8	59.1	1.707	4.392	2.609	2.572	1.528
SD-29	east	147.8	59.1	1.860	4.392	2.609	2.361	1.402
SD-29	west	147.8	59.1	1.860	4.392	2.609	2.361	1.402
SD-29	north	147.8	59.1	1.707	4.392	2.609	2.572	1.528
SD-29	south	147.8	59.1	1.707	4.392	2.609	2.572	1.528
SD-30	east	147.8	59.1	1.860	4.392	2.609	2.361	1.402
SD-30	west	147.8	59.1	1.860	4.392	2.609	2.361	1.402
SD-30	north	147.8	59.1	1.707	4.392	2.609	2.572	1.528
SD-30	south	147.8	59.1	1.707	4.392	2.609	2.572	1.528

ID	Direction	Max L/s	Min L/s	Input L m	Max throw m	Min throw m	Max T/L	Min T/L
SD-31	east	147.8	59.1	1.860	4.392	2.609	2.361	1.402
SD-31	west	147.8	59.1	1.860	4.392	2.609	2.361	1.402
SD-31	north	147.8	59.1	1.707	4.392	2.609	2.572	1.528
SD-31	south	147.8	59.1	1.707	4.392	2.609	2.572	1.528
SD-32	east	163.9	65.5	2.264	4.586	2.881	2.026	1.273
SD-32	west	163.9	65.5	1.860	4.586	2.881	2.466	1.549
SD-32	north	163.9	65.5	1.707	4.586	2.881	2.686	1.687
SD-32	south	163.9	65.5	1.707	4.586	2.881	2.686	1.687
SD-33	east	159.3	63.7	1.745	4.531	2.803	2.597	1.607
SD-33	west	159.3	63.7	2.264	4.531	2.803	2.001	1.238
SD-33	north	159.3	63.7	1.707	4.531	2.803	2.653	1.642
SD-33	south	159.3	63.7	1.707	4.531	2.803	2.653	1.642
SD-34	east	138.6	55.5	1.745	4.281	2.453	2.454	1.406
SD-34	west	138.6	55.5	1.745	4.281	2.453	2.454	1.406
SD-34	north	138.6	55.5	1.707	4.281	2.453	2.507	1.437
SD-34	south	138.6	55.5	1.707	4.281	2.453	2.507	1.437
SD-35	east	138.6	55.5	1.745	4.281	2.453	2.454	1.406
SD-35	west	138.6	55.5	1.745	4.281	2.453	2.454	1.406
SD-35	north	138.6	55.5	1.707	4.281	2.453	2.507	1.437
SD-35	south	138.6	55.5	1.707	4.281	2.453	2.507	1.437
SD-36	east	138.6	55.5	1.745	4.281	2.453	2.454	1.406
SD-36	west	138.6	55.5	1.745	4.281	2.453	2.454	1.406
SD-36	north	138.6	55.5	1.707	4.281	2.453	2.507	1.437
SD-36	south	138.6	55.5	1.707	4.281	2.453	2.507	1.437
SD-37	east	161.4	64.6	1.860	4.556	2.839	2.450	1.526
SD-37	west	161.4	64.6	1.860	4.556	2.839	2.450	1.526

ID	Direction	Max L/s	Min L/s	Input L m	Max throw m	Min throw m	Max T/L	Min T/L
SD-37	north	161.4	64.6	2.021	4.556	2.839	2.255	1.405
SD-37	south	161.4	64.6	1.707	4.556	2.839	2.668	1.663
SD-38	east	161.4	64.6	1.860	4.556	2.839	2.450	1.526
SD-38	west	161.4	64.6	1.860	4.556	2.839	2.450	1.526
SD-38	north	161.4	64.6	2.021	4.556	2.839	2.255	1.405
SD-38	south	161.4	64.6	1.707	4.556	2.839	2.668	1.663
SD-39	east	161.4	64.6	1.860	4.556	2.839	2.450	1.526
SD-39	west	161.4	64.6	1.860	4.556	2.839	2.450	1.526
SD-39	north	161.4	64.6	2.021	4.556	2.839	2.255	1.405
SD-39	south	161.4	64.6	1.707	4.556	2.839	2.668	1.663
SD-40	east	161.4	64.6	1.860	4.556	2.839	2.450	1.526
SD-40	west	161.4	64.6	1.860	4.556	2.839	2.450	1.526
SD-40	north	161.4	64.6	2.021	4.556	2.839	2.255	1.405
SD-40	south	161.4	64.6	1.707	4.556	2.839	2.668	1.663
SD-41	east	161.4	64.6	1.860	4.556	2.839	2.450	1.526
SD-41	west	161.4	64.6	1.860	4.556	2.839	2.450	1.526
SD-41	north	161.4	64.6	2.021	4.556	2.839	2.255	1.405
SD-41	south	161.4	64.6	1.707	4.556	2.839	2.668	1.663
SD-42	east	161.4	64.6	1.860	4.556	2.839	2.450	1.526
SD-42	west	161.4	64.6	1.860	4.556	2.839	2.450	1.526
SD-42	north	161.4	64.6	2.021	4.556	2.839	2.255	1.405
SD-42	south	161.4	64.6	1.707	4.556	2.839	2.668	1.663
SD-43	east	161.4	64.6	1.860	4.556	2.839	2.450	1.526
SD-43	west	161.4	64.6	1.860	4.556	2.839	2.450	1.526
SD-43	north	161.4	64.6	2.021	4.556	2.839	2.255	1.405
SD-43	south	161.4	64.6	1.707	4.556	2.839	2.668	1.663

ID	Direction	Max L/s	Min L/s	Input L m	Max throw m	Min throw m	Max T/L	Min T/L
SD-44	east	178.9	71.6	2.264	4.811	3.018	2.125	1.333
SD-44	west	178.9	71.6	1.860	4.811	3.018	2.586	1.623
SD-44	north	178.9	71.6	2.021	4.811	3.018	2.380	1.493
SD-44	south	178.9	71.6	1.707	4.811	3.018	2.817	1.768
SD-45	east	173.9	69.6	1.745	4.735	2.976	2.714	1.706
SD-45	west	173.9	69.6	2.264	4.735	2.976	2.091	1.314
SD-45	north	173.9	69.6	2.021	4.735	2.976	2.343	1.472
SD-45	south	173.9	69.6	1.707	4.735	2.976	2.773	1.743
SD-46	east	151.4	60.6	1.745	4.435	2.669	2.542	1.530
SD-46	west	151.4	60.6	1.745	4.435	2.669	2.542	1.530
SD-46	north	151.4	60.6	2.021	4.435	2.669	2.194	1.321
SD-46	south	151.4	60.6	1.707	4.435	2.669	2.597	1.563
SD-47	east	151.4	60.6	1.745	4.435	2.669	2.542	1.530
SD-47	west	151.4	60.6	1.745	4.435	2.669	2.542	1.530
SD-47	north	151.4	60.6	2.021	4.435	2.669	2.194	1.321
SD-47	south	151.4	60.6	1.707	4.435	2.669	2.597	1.563
SD-48	east	151.4	60.6	1.745	4.435	2.669	2.542	1.530
SD-48	west	151.4	60.6	1.745	4.435	2.669	2.542	1.530
SD-48	north	151.4	60.6	2.021	4.435	2.669	2.194	1.321
SD-48	south	151.4	60.6	1.707	4.435	2.669	2.597	1.563
SD-49	east	135.5	54.2	1.860	4.242	2.399	2.281	1.290
SD-49	west	135.5	54.2	1.860	4.242	2.399	2.281	1.290
SD-49	north	135.5	54.2	1.394	4.242	2.399	3.043	1.721
SD-49	south	135.5	54.2	2.021	4.242	2.399	2.099	1.187
SD-50	east	147.8	59.1	1.860	4.392	2.609	2.361	1.402
SD-50	west	147.8	59.1	1.860	4.392	2.609	2.361	1.402

ID	Direction	Max L/s	Min L/s	Input L m	Max throw m	Min throw m	Max T/L	Min T/L
SD-50	north	147.8	59.1	1.394	4.392	2.609	3.150	1.871
SD-50	south	147.8	59.1	2.021	4.392	2.609	2.173	1.291
SD-51	east	147.8	59.1	1.860	4.392	2.609	2.361	1.402
SD-51	west	147.8	59.1	1.860	4.392	2.609	2.361	1.402
SD-51	north	147.8	59.1	1.394	4.392	2.609	3.150	1.871
SD-51	south	147.8	59.1	2.021	4.392	2.609	2.173	1.291
SD-52	east	139.7	55.9	1.860	4.293	2.470	2.308	1.328
SD-52	west	139.7	55.9	1.860	4.293	2.470	2.308	1.328
SD-52	north	139.7	55.9	1.394	4.293	2.470	3.080	1.772
SD-52	south	139.7	55.9	2.021	4.293	2.470	2.124	1.222
SD-53	east	141.2	56.5	1.860	4.312	2.497	2.318	1.342
SD-53	west	141.2	56.5	1.860	4.312	2.497	2.318	1.342
SD-53	north	141.2	56.5	1.394	4.312	2.497	3.093	1.791
SD-53	south	141.2	56.5	2.021	4.312	2.497	2.134	1.235
SD-54	east	147.8	59.1	1.860	4.392	2.609	2.361	1.402
SD-54	west	147.8	59.1	1.860	4.392	2.609	2.361	1.402
SD-54	north	147.8	59.1	1.394	4.392	2.609	3.150	1.871
SD-54	south	147.8	59.1	2.021	4.392	2.609	2.173	1.291
SD-55	east	147.8	59.1	1.860	4.392	2.609	2.361	1.402
SD-55	west	147.8	59.1	1.860	4.392	2.609	2.361	1.402
SD-55	north	147.8	59.1	1.394	4.392	2.609	3.150	1.871
SD-55	south	147.8	59.1	2.021	4.392	2.609	2.173	1.291
SD-56	east	155.9	62.4	2.264	4.490	2.746	1.983	1.213
SD-56	west	155.9	62.4	1.860	4.490	2.746	2.414	1.477
SD-56	north	155.9	62.4	1.394	4.490	2.746	3.221	1.970
SD-56	south	155.9	62.4	2.021	4.490	2.746	2.222	1.359

ID	Direction	Max L/s	Min L/s	Input L m	Max throw m	Min throw m	Max T/L	Min T/L
SD-57	east	152.5	61.0	1.745	4.448	2.688	2.550	1.541
SD-57	west	152.5	61.0	2.264	4.448	2.688	1.965	1.187
SD-57	north	152.5	61.0	1.394	4.448	2.688	3.191	1.928
SD-57	south	152.5	61.0	2.021	4.448	2.688	2.201	1.330
SD-58	east	138.6	55.5	1.745	4.281	2.453	2.454	1.406
SD-58	west	138.6	55.5	1.745	4.281	2.453	2.454	1.406
SD-58	north	138.6	55.5	1.394	4.281	2.453	3.071	1.760
SD-58	south	138.6	55.5	2.021	4.281	2.453	2.118	1.214
SD-59	east	138.6	55.5	1.745	4.281	2.453	2.454	1.406
SD-59	west	138.6	55.5	1.745	4.281	2.453	2.454	1.406
SD-59	north	138.6	55.5	1.394	4.281	2.453	3.071	1.760
SD-59	south	138.6	55.5	2.021	4.281	2.453	2.118	1.214
SD-60	east	126.3	50.5	1.745	4.096	2.243	2.348	1.286
SD-60	west	126.3	50.5	1.745	4.096	2.243	2.348	1.286
SD-60	north	126.3	50.5	1.394	4.096	2.243	2.938	1.609
SD-60	south	126.3	50.5	2.021	4.096	2.243	2.027	1.110
SD-61	east	134.3	53.7	1.860	4.227	2.378	2.273	1.279
SD-61	west	134.3	53.7	1.860	4.227	2.378	2.273	1.279
SD-61	north	134.3	53.7	1.707	4.227	2.378	2.476	1.393
SD-61	south	134.3	53.7	1.394	4.227	2.378	3.032	1.706
SD-62	east	134.3	53.7	1.860	4.227	2.378	2.273	1.279
SD-62	west	134.3	53.7	1.860	4.227	2.378	2.273	1.279
SD-62	north	134.3	53.7	1.707	4.227	2.378	2.476	1.393
SD-62	south	134.3	53.7	1.394	4.227	2.378	3.032	1.706
SD-63	east	134.3	53.7	1.860	4.227	2.378	2.273	1.279
SD-63	west	134.3	53.7	1.860	4.227	2.378	2.273	1.279

ID	Direction	Max L/s	Min L/s	Input L m	Max throw m	Min throw m	Max T/L	Min T/L
SD-63	north	134.3	53.7	1.707	4.227	2.378	2.476	1.393
SD-63	south	134.3	53.7	1.394	4.227	2.378	3.032	1.706
SD-64	east	134.3	53.7	1.860	4.227	2.378	2.273	1.279
SD-64	west	134.3	53.7	1.860	4.227	2.378	2.273	1.279
SD-64	north	134.3	53.7	1.707	4.227	2.378	2.476	1.393
SD-64	south	134.3	53.7	1.394	4.227	2.378	3.032	1.706
SD-65	east	134.3	53.7	1.860	4.227	2.378	2.273	1.279
SD-65	west	134.3	53.7	1.860	4.227	2.378	2.273	1.279
SD-65	north	134.3	53.7	1.707	4.227	2.378	2.476	1.393
SD-65	south	134.3	53.7	1.394	4.227	2.378	3.032	1.706
SD-66	east	134.3	53.7	1.860	4.227	2.378	2.273	1.279
SD-66	west	134.3	53.7	1.860	4.227	2.378	2.273	1.279
SD-66	north	134.3	53.7	1.707	4.227	2.378	2.476	1.393
SD-66	south	134.3	53.7	1.394	4.227	2.378	3.032	1.706
SD-67	east	134.3	53.7	1.860	4.227	2.378	2.273	1.279
SD-67	west	134.3	53.7	1.860	4.227	2.378	2.273	1.279
SD-67	north	134.3	53.7	1.707	4.227	2.378	2.476	1.393
SD-67	south	134.3	53.7	1.394	4.227	2.378	3.032	1.706
SD-68	east	148.8	59.5	2.264	4.404	2.626	1.945	1.160
SD-68	west	148.8	59.5	1.860	4.404	2.626	2.368	1.412
SD-68	north	148.8	59.5	1.707	4.404	2.626	2.579	1.538
SD-68	south	148.8	59.5	1.394	4.404	2.626	3.159	1.883
SD-69	east	156.3	62.5	1.745	4.495	2.753	2.577	1.578
SD-69	west	156.3	62.5	2.264	4.495	2.753	1.985	1.216
SD-69	north	156.3	62.5	1.995	4.495	2.753	2.253	1.380
SD-69	south	156.3	62.5	1.394	4.495	2.753	3.224	1.975

ID	Direction	Max L/s	Min L/s	Input L m	Max throw m	Min throw m	Max T/L	Min T/L
SD-70	east	137.6	55.0	1.745	4.268	2.435	2.446	1.396
SD-70	west	137.6	55.0	1.745	4.268	2.435	2.446	1.396
SD-70	north	137.6	55.0	1.995	4.268	2.435	2.139	1.221
SD-70	south	137.6	55.0	1.394	4.268	2.435	3.061	1.747
SD-71	east	137.6	55.0	1.745	4.268	2.435	2.446	1.396
SD-71	west	137.6	55.0	1.745	4.268	2.435	2.446	1.396
SD-71	north	137.6	55.0	1.995	4.268	2.435	2.139	1.221
SD-71	south	137.6	55.0	1.394	4.268	2.435	3.061	1.747
SD-72	east	137.6	55.0	1.745	4.268	2.435	2.446	1.396
SD-72	west	137.6	55.0	1.745	4.268	2.435	2.446	1.396
SD-72	north	137.6	55.0	1.995	4.268	2.435	2.139	1.221
SD-72	south	137.6	55.0	1.394	4.268	2.435	3.061	1.747
SD-73	east	173.7	69.5	1.745	4.731	2.974	2.712	1.704
SD-73	west	173.7	69.5	1.745	4.731	2.974	2.712	1.704
SD-73	north	173.7	69.5	2.282	4.731	2.974	2.073	1.303
SD-73	south	173.7	69.5	1.995	4.731	2.974	2.372	1.490
SD-74	east	173.7	69.5	1.745	4.731	2.974	2.712	1.704
SD-74	west	173.7	69.5	1.745	4.731	2.974	2.712	1.704
SD-74	north	173.7	69.5	2.282	4.731	2.974	2.073	1.303
SD-74	south	173.7	69.5	1.995	4.731	2.974	2.372	1.490
SD-75	east	173.7	69.5	1.745	4.731	2.974	2.712	1.704
SD-75	west	173.7	69.5	1.745	4.731	2.974	2.712	1.704
SD-75	north	173.7	69.5	2.282	4.731	2.974	2.073	1.303
SD-75	south	173.7	69.5	1.995	4.731	2.974	2.372	1.490
SD-76	east	173.7	69.5	1.745	4.731	2.974	2.712	1.704
SD-76	west	173.7	69.5	1.745	4.731	2.974	2.712	1.704

ID	Direction	Max L/s	Min L/s	Input L m	Max throw m	Min throw m	Max T/L	Min T/L
SD-76	north	173.7	69.5	2.282	4.731	2.974	2.073	1.303
SD-76	south	173.7	69.5	1.995	4.731	2.974	2.372	1.490
SD-77	east	185.3	74.1	1.745	4.908	3.073	2.813	1.761
SD-77	west	185.3	74.1	1.745	4.908	3.073	2.813	1.761
SD-77	north	185.3	74.1	2.282	4.908	3.073	2.150	1.346
SD-77	south	185.3	74.1	2.282	4.908	3.073	2.150	1.346
SD-78	east	185.3	74.1	1.745	4.908	3.073	2.813	1.761
SD-78	west	185.3	74.1	1.745	4.908	3.073	2.813	1.761
SD-78	north	185.3	74.1	2.282	4.908	3.073	2.150	1.346
SD-78	south	185.3	74.1	2.282	4.908	3.073	2.150	1.346
SD-79	east	185.3	74.1	1.745	4.908	3.073	2.813	1.761
SD-79	west	185.3	74.1	1.745	4.908	3.073	2.813	1.761
SD-79	north	185.3	74.1	2.282	4.908	3.073	2.150	1.346
SD-79	south	185.3	74.1	2.282	4.908	3.073	2.150	1.346
SD-80	east	169.9	68.0	1.745	4.675	2.942	2.680	1.686
SD-80	west	169.9	68.0	1.745	4.675	2.942	2.680	1.686
SD-80	north	169.9	68.0	2.282	4.675	2.942	2.048	1.289
SD-80	south	169.9	68.0	2.282	4.675	2.942	2.048	1.289

Leaf group flows

Group	Max L/s	Min L/s
VAV-01	575.7	230.3
VAV-02	584.4	233.8
VAV-03	585.8	234.3
VAV-04	616.7	246.7
VAV-05	590.2	236.1
VAV-06	539.0	215.6
VAV-07	618.4	247.4
VAV-08	618.4	247.4
VAV-09	618.4	247.4
VAV-10	652.0	260.8
VAV-11	623.2	249.3
VAV-12	580.1	232.0
VAV-13	551.8	220.7
VAV-14	556.0	222.4
VAV-15	557.6	223.0
VAV-16	586.9	234.7
VAV-17	585.1	234.0
VAV-18	540.1	216.0
VAV-19	718.0	287.2
VAV-20	702.6	281.0

Selected duct sections

Original Canvas lecture p7: office supply main/branch maxima 10/8 m/s; recommended low-velocity gradient 0.65–1.23 Pa/m, using 1.23 as selection cap. p8 preferred aspect 4:1. p9 chart conditions density1.20 kg/m3 and roughness0.09 mm. Finite dimension grid and viscosity1.81e-5 Pa s remain design assumptions. No lower pressure-gradient rejection is imposed because the recommended band is not a required minimum.

Section	L/s	Size mm	m/s	Pa/m
A-01	12000.4	1350 x 900	9.88	0.752
A-02	12000.4	1350 x 900	9.88	0.752
A-03	12000.4	1350 x 900	9.88	0.752
D-01	10579.7	1250 x 850	9.96	0.825
D-02	1665.4	500 x 450	7.40	1.171
D-03	1107.8	500 x 350	6.33	1.036
D-04	551.8	400 x 250	5.52	1.144
D-05	1712.1	600 x 400	7.13	1.076
D-06	1125.2	500 x 350	6.43	1.067
D-07	540.1	400 x 250	5.40	1.099
D-08	7202.3	850 x 850	9.97	1.017
D-09	1855.3	550 x 450	7.50	1.138
D-10	1236.9	450 x 400	6.87	1.167
D-11	618.4	350 x 300	5.89	1.216
D-12	1855.3	550 x 450	7.50	1.138
D-13	1203.3	500 x 350	6.88	1.209
D-14	580.1	350 x 300	5.52	1.080
D-15	3491.8	650 x 600	8.95	1.202
D-16	1745.9	600 x 400	7.27	1.117
D-17	1160.1	500 x 350	6.63	1.129
D-18	575.7	350 x 300	5.48	1.065
D-19	1745.9	600 x 400	7.27	1.117

Section	L/s	Size mm	m/s	Pa/m
D-20	1129.1	500 x 350	6.45	1.074
D-21	539.0	400 x 250	5.39	1.095
D-22	1420.6	500 x 400	7.10	1.171
D-23	1420.6	500 x 400	7.10	1.171
D-24	702.6	400 x 300	5.86	1.119
D-25	575.7	350 x 300	5.48	1.065
D-26	584.4	350 x 300	5.57	1.095
D-27	585.8	350 x 300	5.58	1.099
D-28	616.7	350 x 300	5.87	1.210
D-29	590.2	350 x 300	5.62	1.115
D-30	539.0	400 x 250	5.39	1.095
D-31	618.4	350 x 300	5.89	1.216
D-32	618.4	350 x 300	5.89	1.216
D-33	618.4	350 x 300	5.89	1.216
D-34	652.0	450 x 250	5.80	1.188
D-35	623.2	450 x 250	5.54	1.093
D-36	580.1	350 x 300	5.52	1.080
D-37	551.8	400 x 250	5.52	1.144
D-38	556.0	400 x 250	5.56	1.160
D-39	557.6	400 x 250	5.58	1.166
D-40	586.9	350 x 300	5.59	1.103
D-41	585.1	350 x 300	5.57	1.097
D-42	540.1	400 x 250	5.40	1.099
D-43	718.0	400 x 300	5.98	1.165
D-44	702.6	400 x 300	5.86	1.119

Saved terminal connections

Saved native drawing correspondence covers 80 terminals, 20 groups and 47 rigid sections. The longest straight centre-to-centre planar flexible connection is 2.976 m. This does not verify physical ports, installed flexible lengths, labels, three-dimensional coordination or regulatory compliance.

Complete source requirement matrix

ID	Source requirement	Execution	Acceptance state	Acceptance criterion
R001-identity	The Hong Kong Polytechnic University Department of Building Environment and Energy Engineering	pending	pending review	Correct project identity and group size;no internal iteration labels in final deliverables.; full source quote remains authoritative, acceptance pending
R002-identity	BSE3227 HVACR 1 2026/27 Semester 1	pending	pending review	Correct project identity and group size;no internal iteration labels in final deliverables.; full source quote remains authoritative, acceptance pending
R003-identity	Mini-design assignment	pending	pending review	Correct project identity and group size;no internal iteration labels in final deliverables.; full source quote remains authoritative, acceptance pending
R004-identity	Students should form groups of up to five members, with each group tasked with designing the air conditioning layout for an open-plan office on a typical floor. The architectural layout plan for this floor is provided. It is assumed that a variable air volume (VAV) system will be utilized in the design.	pending	pending review	Correct project identity and group size;no internal iteration labels in final deliverables.; full source quote remains authoritative, acceptance pending
R004-layout	Students should form groups of up to five members, with each group tasked with designing the air conditioning layout for an open-plan office on a typical floor. The architectural layout plan for this floor is provided. It is assumed that a variable air volume (VAV) system will be utilized in the design.	pending	pending review	Source-bound terminal positions,flow conservation,L definition and graphical correspondence;keep design choices distinct from measured facts.; full source quote remains authoritative, acceptance pending
R004-routing	Students should form groups of up to five members, with each group tasked with designing the air conditioning layout for an open-plan office on a typical floor. The architectural layout plan for this floor is provided. It is assumed that a variable air volume (VAV) system will be utilized in the design.	pending	pending review	All diffusers connected;4to5outlets per VAV;flex within5m;straight routes/minimal turns and distribution examined.; full source quote remains authoritative, acceptance pending
R005-delivery	Each group is required to complete the following tasks.	pending	pending review	Independent DWG reopen,actual plot review,complete original requirement coverage and explicit limitations;no portal submission.; full source quote remains authoritative, acceptance pending
R006-load	1. Space air diffusion design for maximum load condition	pending	pending review	Independent thermal/airflow arithmetic;given area versus interpreted geometry explicit;no actual room-load or electrical-power claim.; full source quote remains authoritative, acceptance pending

ID	Source requirement	Execution	Acceptance state	Acceptance criterion
R006-layout	1. Space air diffusion design for maximum load condition	pending	pending review	Source-bound terminal positions,flow conservation,L definition and graphical correspondence;keep design choices distinct from measured facts.; full source quote remains authoritative, acceptance pending
R006-terminal	1. Space air diffusion design for maximum load condition	pending	pending review	Source-reviewed neck/panel configuration,proper throw column and bounded interpolation for every case;out-of-catalogue remains explicit.; full source quote remains authoritative, acceptance pending
R006-adpi	1. Space air diffusion design for maximum load condition	pending	pending review	Retain selected edition/category and source domains;no ADPI pass from ratio alone,clamping or missing facts.; full source quote remains authoritative, acceptance pending
R007-load	• Calculate the cooling load of the open plan office using rule of thumb. • Determine the total supply air flow rate of the office. • Distribute supply air diffusers on the layout plan. • Calculate the supply air flow rate of each diffuser. • Find out the characteristic room length, L. • Refer to the performance data of KRUEGER “Louvered Face Diffusers 1400, 51400 Series 4-cone”, and select a suitable type of diffuser. • From the diffuser performance data, find the corresponding T0.25. • Calculate T0.25 / L. • Check if it meets the ASHRAE ADPI requirement.	pending	pending review	Independent thermal/airflow arithmetic;given area versus interpreted geometry explicit;no actual room-load or electrical-power claim.; full source quote remains authoritative, acceptance pending
R007-layout	• Calculate the cooling load of the open plan office using rule of thumb. • Determine the total supply air flow rate of the office. • Distribute supply air diffusers on the layout plan. • Calculate the supply air flow rate of each diffuser. • Find out the characteristic room length, L. • Refer to the performance data of KRUEGER “Louvered Face Diffusers 1400, 51400 Series 4-cone”, and select a suitable type of diffuser. • From the diffuser performance data, find the corresponding T0.25. • Calculate T0.25 / L. • Check if it meets the ASHRAE ADPI requirement.	pending	pending review	Source-bound terminal positions,flow conservation,L definition and graphical correspondence;keep design choices distinct from measured facts.; full source quote remains authoritative, acceptance pending
R007-terminal	• Calculate the cooling load of the open plan office using rule of thumb. • Determine the total supply air flow rate of the office. • Distribute supply air diffusers on the layout plan. • Calculate the supply air flow rate of each diffuser. • Find out the characteristic room length, L. • Refer to the performance data of KRUEGER “Louvered Face Diffusers 1400, 51400 Series 4-cone”, and select a suitable type of diffuser. • From the diffuser performance data, find the corresponding T0.25. • Calculate T0.25 / L. • Check if it meets the ASHRAE ADPI requirement.	pending	pending review	Source-reviewed neck/panel configuration,proper throw column and bounded interpolation for every case;out-of-catalogue remains explicit.; full source quote remains authoritative, acceptance pending

ID	Source requirement	Execution	Acceptance state	Acceptance criterion
R007-adpi	• Calculate the cooling load of the open plan office using rule of thumb. • Determine the total supply air flow rate of the office. • Distribute supply air diffusers on the layout plan. • Calculate the supply air flow rate of each diffuser. • Find out the characteristic room length, L. • Refer to the performance data of KRUEGER “Louvered Face Diffusers 1400, 51400 Series 4-cone”, and select a suitable type of diffuser. • From the diffuser performance data, find the corresponding T0.25. • Calculate T0.25 / L. • Check if it meets the ASHRAE ADPI requirement.	pending	pending review	Retain selected edition/category and source domains;no ADPI pass from ratio alone,clamping or missing facts.; full source quote remains authoritative, acceptance pending
R008-minimum	2. Space air diffusion design for minimum load condition	pending	pending review	Every original terminal retains identity;minimum flow equals0.4maximum with explicit units.; full source quote remains authoritative, acceptance pending
R009-minimum	• Calculate the supply air flow rate of each diffuser with a minimum turn down ratio of 40%. • From the diffuser performance data, find the corresponding T0.25. • Calculate T0.25 / L. • Check if it meets the ASHRAE ADPI requirement.	pending	pending review	Every original terminal retains identity;minimum flow equals0.4maximum with explicit units.; full source quote remains authoritative, acceptance pending
R009-terminal	• Calculate the supply air flow rate of each diffuser with a minimum turn down ratio of 40%. • From the diffuser performance data, find the corresponding T0.25. • Calculate T0.25 / L. • Check if it meets the ASHRAE ADPI requirement.	pending	pending review	Source-reviewed neck/panel configuration,proper throw column and bounded interpolation for every case;out-of-catalogue remains explicit.; full source quote remains authoritative, acceptance pending
R009-adpi	• Calculate the supply air flow rate of each diffuser with a minimum turn down ratio of 40%. • From the diffuser performance data, find the corresponding T0.25. • Calculate T0.25 / L. • Check if it meets the ASHRAE ADPI requirement.	pending	pending review	Retain selected edition/category and source domains;no ADPI pass from ratio alone,clamping or missing facts.; full source quote remains authoritative, acceptance pending
R010-routing	3. Single-line diagram of AC layout plan	pending	pending review	All diffusers connected;4to5outlets per VAV;flex within5m;straight routes/minimal turns and distribution examined.; full source quote remains authoritative, acceptance pending
R011-routing	• Use single lines to sketch the air duct layout for distribution of supply air to all supply air diffusers in the office. • Use VAV boxes with four to five outlets, but limit the flexible duct length (from the VAV box to the diffuser) to be within 5 m. • Use straight ductwork as far as possible. • Minimize bends and turns. • Plan branch ducts to extend from the main trunk to distribute air evenly throughout the space.	pending	pending review	All diffusers connected;4to5outlets per VAV;flex within5m;straight routes/minimal turns and distribution examined.; full source quote remains authoritative, acceptance pending

ID	Source requirement	Execution	Acceptance state	Acceptance criterion
R011-delivery	• Use single lines to sketch the air duct layout for distribution of supply air to all supply air diffusers in the office. • Use VAV boxes with four to five outlets, but limit the flexible duct length (from the VAV box to the diffuser) to be within 5 m. • Use straight ductwork as far as possible. • Minimize bends and turns. • Plan branch ducts to extend from the main trunk to distribute air evenly throughout the space.	pending	pending review	Independent DWG reopen,actual plot review,complete original requirement coverage and explicit limitations;no portal submission.; full source quote remains authoritative, acceptance pending
R012-sizing	4. Sizing of supply air ducts	pending	pending review	All sections retain IDs/flows;source method/criteria and aspect ratio preference are explicit;equation audit is separate from chart/method applicability.; full source quote remains authoritative, acceptance pending
R013-sizing	• Determine the size of the main air ducts and each section of the branch air ducts. • The air duct size should satisfy the requirements of maximum air velocity and limiting air-pressure drops. • Use the Friction Chart in ASHRAE Fundamentals or Online Ductular. • The maximum aspect ratio of rectangular air duct is preferred to be 4:1. • Label each section of the air duct and tabulate the air duct sizing calculation in a table. • The following table for air duct sizing is suggested for your use.	pending	pending review	All sections retain IDs/flows;source method/criteria and aspect ratio preference are explicit;equation audit is separate from chart/method applicability.; full source quote remains authoritative, acceptance pending
R014-schedule	Section Volume flow rate (L/s) Rectangular duct size (mm×mm) Air velocity (m/s) Pressure drop (Pa/m)	pending	pending review	Every section appears withL/s,mm by mm,m/s andPa/m,consistent with editable drawing and calculations.; full source quote remains authoritative, acceptance pending
R014-delivery	Section Volume flow rate (L/s) Rectangular duct size (mm×mm) Air velocity (m/s) Pressure drop (Pa/m)	pending	pending review	Independent DWG reopen,actual plot review,complete original requirement coverage and explicit limitations;no portal submission.; full source quote remains authoritative, acceptance pending
producer-air-allocation-P007	• Calculate the cooling load of the open plan office using rule of thumb. • Determine the total supply air flow rate of the office. • Distribute supply air diffusers on the layout plan. • Calculate the supply air flow rate of each diffuser. • Find out the characteristic room length, L. • Refer to the performance data of KRUEGER “Louvered Face Diffusers 1400, 51400 Series 4-cone”, and select a suitable type of diffuser. • From the diffuser performance data, find the corresponding T0.25. • Calculate T0.25 / L. • Check if it meets the ASHRAE ADPI requirement.	pass	pending review	Execute classified preliminary thermal/airflow balance and nearest-centre scenario allocation;actual room load and physical layout acceptance remain pending.

ID	Source requirement	Execution	Acceptance state	Acceptance criterion
producer-terminal-lookup-P007	• Calculate the cooling load of the open plan office using rule of thumb. • Determine the total supply air flow rate of the office. • Distribute supply air diffusers on the layout plan. • Calculate the supply air flow rate of each diffuser. • Find out the characteristic room length, L. • Refer to the performance data of KRUEGER “Louvered Face Diffusers 1400, 51400 Series 4-cone”, and select a suitable type of diffuser. • From the diffuser performance data, find the corresponding T0.25. • Calculate T0.25 / L. • Check if it meets the ASHRAE ADPI requirement.	pass	pending review	Execute source-pinned bounded maximum-flow throw lookup;product,L and ADPI acceptance remain pending.
producer-terminal-lookup-P009	• Calculate the supply air flow rate of each diffuser with a minimum turn down ratio of 40%. • From the diffuser performance data, find the corresponding T0.25. • Calculate T0.25 / L. • Check if it meets the ASHRAE ADPI requirement.	pass	pending review	Execute the same original terminal inventory at40percent minimum flow without extrapolation;ADPI applicability remains pending.
producer-duct-schedule-P013	• Determine the size of the main air ducts and each section of the branch air ducts. • The air duct size should satisfy the requirements of maximum air velocity and limiting air-pressure drops. • Use the Friction Chart in ASHRAE Fundamentals or Online Ductular. • The maximum aspect ratio of rectangular air duct is preferred to be 4:1. • Label each section of the air duct and tabulate the air duct sizing calculation in a table. • The following table for air duct sizing is suggested for your use.	pass	pending review	Execute conditional section numerical sizing under declared source criteria;chart-method,geometry and manufactured-size applicability remain pending.
producer-duct-sheet-P011	• Use single lines to sketch the air duct layout for distribution of supply air to all supply air diffusers in the office. • Use VAV boxes with four to five outlets, but limit the flexible duct length (from the VAV box to the diffuser) to be within 5 m. • Use straight ductwork as far as possible. • Minimize bends and turns. • Plan branch ducts to extend from the main trunk to distribute air evenly throughout the space.	pass	pending review	Bind source section labels/schedule to the preserved single-line design scenario;physical routing/installed lengths remain pending.
producer-native-sheet-P011	• Use single lines to sketch the air duct layout for distribution of supply air to all supply air diffusers in the office. • Use VAV boxes with four to five outlets, but limit the flexible duct length (from the VAV box to the diffuser) to be within 5 m. • Use straight ductwork as far as possible. • Minimize bends and turns. • Plan branch ducts to extend from the main trunk to distribute air evenly throughout the space.	pass	pending review	Save and independently reopen native DWG and plot at declared scale;actual visual and full engineering review remain separate.

ID	Source requirement	Execution	Acceptance state	Acceptance criterion
hk-flex-check	• Use single lines to sketch the air duct layout for distribution of supply air to all supply air diffusers in the office. • Use VAV boxes with four to five outlets, but limit the flexible duct length (from the VAV box to the diffuser) to be within 5 m. • Use straight ductwork as far as possible. • Minimize bends and turns. • Plan branch ducts to extend from the main trunk to distribute air evenly throughout the space.	pending	pending review	Additional Hong Kong project check selected by user; keep assignment 5m criterion distinct from ordinary FSD 4m path; installed evidence and applicability pending

Native files and plots are separate workflow artifacts. Their visual review, physical connections, source applicability and final delivery requirements remain distinct.